

Application Form for Teaching Innovation Grant (TIG) 2026

PART I: SUMMARY OF PROPOSAL

1. Project Title:

Building BRIDGE Between English-medium Medical Training and Chinese Clinical Practice to Enhance Communication Skills and Clinical Reasoning

2. Name(s) of applicant(s) and academic affiliation (*more rows could be added*):

	Name	Post	School
Principal Investigator	Xie Sicong, Cecilia	Lecturer, Course coordinator of <i>MED 4010 Preventive Medicine and General Practice</i>	MED
Co- investigator:	Wang Benyou	Assistant Professor	SDS
Co- investigator:	LAU Mai Wai, Maria	Lecturer, Course coordinator of <i>MED4200: Communication Skills II</i>	MED

3. Expected duration of this project (in year and month)

2 years

Proposed commencement:

September 2026

Estimated completion:

August 2028

4. Category of the proposal

Please restrict your selection to three categories in order of relevancy and give “1” “2” and “3” in the appropriate box(es), with “1” being the most relevant category.

Relevancy	Categories	Relevancy	Categories
	Innovative teaching method		Assessment
2	Courseware development	1	eLearning*
	Experiential learning		Student learning and experience
3	Teaching and learning activities		Outcomes-based approach (OBA)
	Survey research and evaluation		Professional development
	Quality assurance		Others (please specify):

* eLearning includes multimedia learning, technology-enhanced learning.

5. Abstract of the proposal

Medical education at CUHK-Shenzhen needs to bridge the gap between English-medium medical training and Chinese clinical practice. While students study medicine and complete curricular assessments in English, they communicate mainly in Chinese with patients and

clinical preceptors during clinical rotations, creating challenges in bilingual doctor-patient communication, case discussion, and exam preparation.

In response, the project will build BRIDGE (Bilingual Repository for Intelligent Discourse and General Practice Examination), an AI-supported bilingual learning platform for clinical education, in collaboration with Prof. Wang Benyou's team from the School of Data and Science. Faculty and students will jointly transform real, de-identified local clinical cases into a reusable teaching case repository. Based on these expert-reviewed cases, BRIDGE will provide AI-simulated patient (AI-SP), AI-tutor and Obstructive Structured Clinical Examination (OSCE) simulation, enabling students to practice history-taking, patient consultation, and clinical reasoning in both English and Chinese, with immediate and individualized formative feedback.

To be implemented from 2026 to 2028, this project will build BRIDGE platform, featuring at least 100 teaching cases, AI-SP/AI-tutor and OSCE simulation functions, with access provided to all clinical students. Its impact will be assessed through students' OSCE performance, pre- and post-intervention self-efficacy surveys, and focus-group evaluations of user experience, ultimately seeking to strengthen students' bilingual communication skills and clinical reasoning.

PART II DETAILS OF THE PROJECT PROPOSAL

1. The background of proposed innovation and identified T&L Problem

The Clinical Medicine (CMED) curriculum at CUHK-Shenzhen meets international medical education requirements and is delivered in English. However, students rotate in local teaching hospitals, where clinical teaching and learning (T&L) is primarily in Chinese. This creates pressing challenges in bilingual doctor-patient communication, clinical reasoning training, and exam preparation^(1,2).

Existing studies reveals the mismatch between the language of training and practice affects not students' biomedical knowledge, but their competence and confidence to write medical records, communicate with patients and express clinical reasoning⁽²⁻⁶⁾. Case record writing, clinical reasoning practice through structured case-based questioning and OSCE practice with SPs have been shown to improve skills mastery and self-efficacy, particularly when immediate and individualized feedback is provided^(5,7-10).

Unfortunately, such T&L activities remain difficult to implement due to busy clinical work, increasing class size and limited bilingual clinical tutors. Large language model (LLM) based artificial intelligence (AI) may help address this issue through AI-SP and automated formative feedback^(9,11-17). However, existing AI platforms are not fully suited to bilingual clinical education in China due to limited bilingual integration, insufficient adaptation to local clinical contexts, and barriers related to cost or internet access.

Within this context, *MED 4010 Preventive Medicine and General Practice (PMGP)*, a Year 4 core clinical course, has identified three major T&L problems after two years of implementation:

1) Fragmented clinical teaching resources - Valuable local general practice (GP) cases are not systematically converted into structured, reusable teaching resources, making case-based learning (CBL) opportunistic and difficult to scale up.

2) Limited practice opportunities and formative feedback - Students lack opportunities to practice bilingual history-taking, patient consultation, and clinical reasoning discussions, as well as access to personalized formative feedback, resulting in weak professional expression, low confidence, and sub-optimal OSCE performance.

3) Unguided AI use - Students currently use AI for coursework without adequate pedagogical guidance, potentially resulting in flawed clinical reasoning and inappropriate use of medical English^(16,18).

Collectively, these challenges point to an urgent need for an accessible, context-sensitive, and bilingual support system that provides structured opportunities for students to practice history-taking, patient consultation, and clinical reasoning while offering timely and personalized feedback, to better align English-medium medical training with Chinese clinical practice and enhance students' communication skills and clinical reasoning.

2. Proposed Innovative Teaching Method - The BRIDGE Platform

This project proposes to build BRIDGE, an AI-supported bilingual clinical learning platform, to address the above-mentioned T&L problems (Figure 1), through the following key functions:

1) Faculty-student Co-created Teaching Case Repository - Students will convert local clinical cases into English teaching cases using a standardized template and receive GP mentor feedback to enhance case writing skills and clinical reasoning. GP experts will then review cases and create structured Q&As and marking rubrics for AI-SP/AI-tutor, making valuable teaching resources accessible to all clinical students. (Clerking case record is mandatory to all students in PMGP course, under the structure of "4+1 weekly teaching model", in which students rotate in local hospitals from Monday to Thursday and attend GP mentor group meetings on Friday for case discussion and reflection in English.)

2) AI-SP/AI-tutor - Leveraging existing LLM and AI-agent technologies, AI-SP/AI-tutor will be created based on the expert-reviewed cases, case-based Q&As and rubrics, to provide students with unlimited self-directed practice opportunities in bilingual history-taking, patient consultation, and clinical reasoning, while receiving immediate and individualized formative feedback.

3) OSCE simulation - Drawing on expert-reviewed cases, the BRIDGE platform will create OSCE simulations modeled on authentic OSCE formats, enabling students to practice before examinations, build confidence and ultimately demonstrate their full potential in examinations.

The BRIDGE platform will be developed primarily by Prof. Wang Benyou's team, building on its existing model⁽¹⁹⁾ and guided by clinical teaching needs and user feedback.

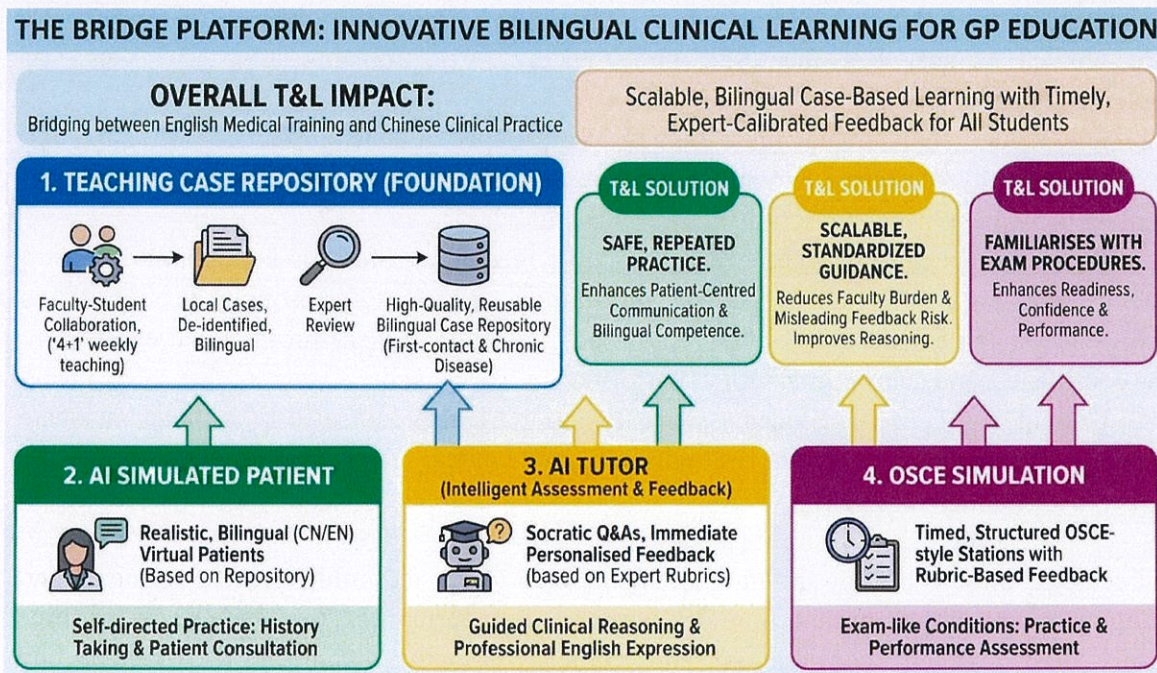


Figure 1. The BRIDGE Platform

3. Implementation plan and timeline:

BRIDGE Strategic Development Plan: 24-Month Roadmap (2026-2028)

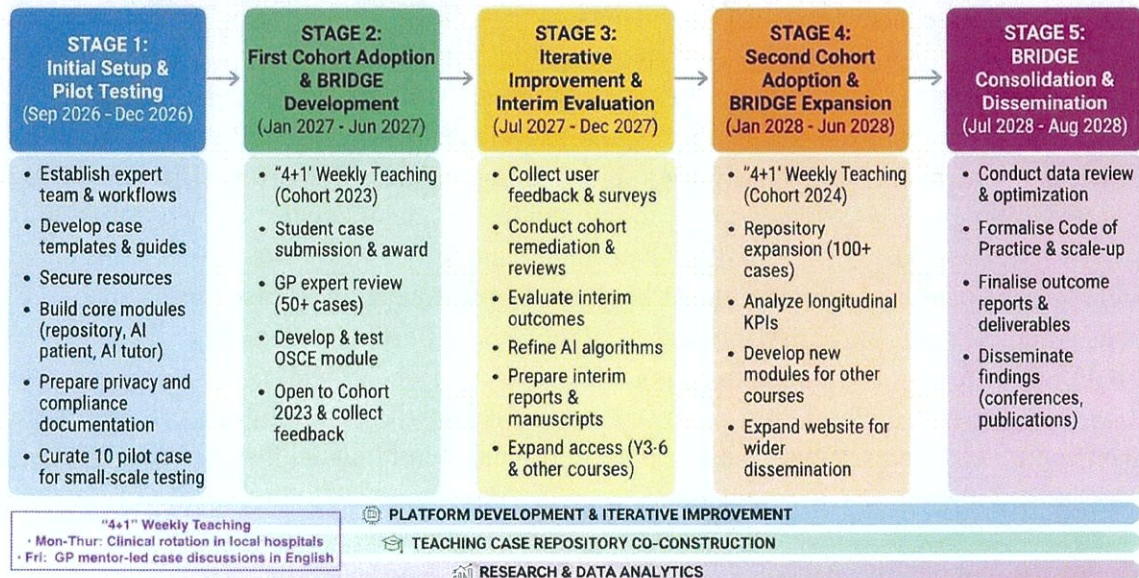


Figure 2. BRIDGE Strategic Development Plan:24-Month Road-map

The BRIDGE development plan is proposed for 24 months (Figure 2), comprising:

- 1) Development of BRIDGE platform, delivered through a web-based interface;
- 2) Teaching case repository of at least 100 cases co-created by students and faculty, reviewed and approved by GP experts;

3) A research study evaluating BRIDGE integration into the PMGP course to enhance bilingual communication skills and clinical reasoning among 2023 and 2024 cohorts.

4. Assessment plan, expected outcomes and deliverables

The project will use a mixed-methods design to evaluate student learning outcomes and the quality of the BRIDGE platform. Quantitative analyses will examine associations between BRIDGE engagement, PMGP OSCE performance and changes in self-efficacy. Qualitative focus groups will explore students' experiences, satisfaction, perceived educational value, usability, barriers and facilitators to engagement, and suggestions for platform improvement.

1) Students' competence will be evaluated using PMGP OSCE performance data from the 2023 cohort (n=53) and 2024 cohort (n=66), with the total OSCE score as the primary outcome and individual station scores as secondary outcomes. Students' active engagement time on BRIDGE will be extracted from the platform back-end and defined as the cumulative time spent on active learning activities, excluding inactive login time. The association between BRIDGE engagement and OSCE performance will be examined by multi-variable regression analysis adjusting for cohort year and relevant baseline academic characteristics.

2) Students' self-efficacy will be evaluated using pre- and post-intervention Likert-scale surveys. The surveys will evaluate students perceived self-efficacy in completing clinical and course-related tasks, including history-taking, patient consultation, case discussion, and OSCEs, in both English and Chinese. Changes in self-efficacy scores will be analyzed using paired statistical methods or mixed-effects models, as appropriate.

3) The quality of BRIDGE will be evaluated through semi-structured focus group interviews conducted 1-2 weeks after students complete all fourth-year courses and examinations. Participants will be purposively sampled to capture diverse experiences across platform engagement levels and academic performance. Interviews will be audio-recorded, transcribed verbatim, and analyzed thematically by two independent researchers. Themes will focus on satisfaction, perceived educational value, usability, barriers and facilitators to engagement, and recommendations for refinement.

Ethical approval will be obtained before study commencement. All clinical cases will be de-identified, and student assessment, survey, and platform data will be anonymized before analysis. Data will be used only for educational evaluation and platform improvement, with access restricted to authorized project members in accordance with institutional ethics and data protection requirements.

Key deliverables include:

- a) The BRIDGE platform with at least 100 expert-reviewed teaching cases, and functions of AI-SP, AI-tutor and OSCE simulation;
- b) Supporting teaching package with standardized teaching case template, structured case discussion guide and "4+1 weekly teaching model", with scalability across clinical courses in CMED programs;
- c) Scholarly outputs & dissemination including presentation at local or international

medical education conferences and at least one publication of high-quality educational research paper on index journals;

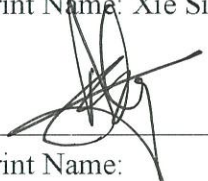
d) Broader educational impact to explore adaptation of BRIDGE platform and the bilingual clinical teaching model to other clinical courses in CUHK-Shenzhen, and further to bilingual CMED programs in China and globally.

PART III BUDGET

Item	Amount (RMB)			Justification
	2026-27	2027-28	Total	
(a) Labor cost				
(i) Student helper	50/person/hr. x 250 hours = 12500	50/person/hr. x 150 hours = 7500	20000	- To build up, refine and maintain the BRIDGE platform according to teaching needs and user feedback. - To assist in research studies. - Co-authorship of the teaching cases.
(b) Travel & Conference				
(i) Registration fee and overseas travel expenditure	5000	10000	15000	- To disseminate the project outcome and exchange the new insights of the medical training in AI era.
(c) Consultation fee				
(i) Completion Evaluation Expert		5000	5000	- External experts to review the project deliverables, outcomes, final report, and provide written evaluation comments.
(b) General Expenses				
(i) Service charge	5000	5000	10000	- Subscription fee of accessible LLM and other services like voice recognition, website deployment and data storage.
(ii) Publication fee	15000	15000	30000	- To target PMC/Open Access publications on index journals.
Total	37500	42500	80000	

PART IV Endorsement

Signature of Principal Investigator:  Date: 2026.6.7
(Print Name: Xie Sicong, Cecilia 谢思聪)

Endorsed by Dean:  Date: _____
(Print Name: _____)

References:

1. Mikkonen K, Elo S, Kuivila HM, Tuomikoski AM, Kääriäinen M. Culturally and linguistically diverse healthcare students' experiences of learning in a clinical environment: a systematic review of qualitative studies. *Int J Nurs Stud*. 2016 Feb 1; Language and Communication Issues in Health Care 54:173–87. doi:10.1016/j.ijnurstu.2015.06.004
2. Tsui JJ, Li EY, Xie CS, Zhu Z, Yin X, Tsui BCH, et al. Bridging language gaps: a study of linguistic transfer perceptions among Chinese medical students in an English-medium instruction program. *Front Med*. 2026 May 4;13. doi:10.3389/fmed.2026.1798909
3. Abi Raad V, Raad K, Daaboul Y, Korjian S, Asmar N, Jammal M, et al. Medical education in a foreign language and history-taking in the native language in Lebanon – a nationwide survey. *BMC Med Educ*. 2016 Nov 22;16(1):298. doi:10.1186/s12909-016-0826-7
4. Li D, Zheng J, Han Y. Unpacking translanguaging practices in intercultural healthcare communication: experiences of international medical students in a Chinese hospital. *Appl Linguist*. 2025 Nov 20; amaf077. doi:10.1093/applin/amaf077
5. Zou L, Su J, Li J, Wang J, Kang J, Yin A, et al. Application of bilingual simulated patients in the medical history collection for international medical students in China. *BMC Med Educ*. 2023 Jul 22;23(1):525. doi:10.1186/s12909-023-04480-1
6. Hamad AA, Mustaffa DB, Alnajjar AZ, Amro R, Deameh MG, Amin B, et al. Decolonizing medical education: a systematic review of educational language barriers in countries using foreign languages for instruction. *BMC Med Educ*. 2025 May 13;25(1):701. doi:10.1186/s12909-025-07251-2
7. Gonullu I, Bayazit A, Erden S. Exploring medical students' perceptions of individual and group-based clinical reasoning with virtual patients: a qualitative study. *BMC Med Educ*. 2024 Feb 25;24(1):189. doi:10.1186/s12909-024-05121-x
8. Vest BM, Lynch A, McGuigan D, Servoss T, Zinnerstrom K, Symons AB. Using standardized patient encounters to teach longitudinal continuity of care in a family medicine clerkship. *BMC Med Educ*. 2016 Aug 17;16(1):208. doi:10.1186/s12909-016-0733-y
9. Ali M, Han SC, Bilal HSM, Lee S, Kang MJY, Kang BH, et al. iCBLS: An interactive case-based learning system for medical education. *Int J Med Inf*. 2018 Jan;109:55–69. doi:10.1016/j.ijmedinf.2017.11.004
10. Moghadami M, Amini M, Moghadami M, Dalal B, Et. A. Teaching clinical reasoning to undergraduate medical students by illness script method: a randomized controlled trial. *BMC Med Educ*. 2021. doi:10.1186/s12909-021-02522-0
11. Deschênes MF, Fernandez N, Lechasseur K, Caty MÈ, Uctu BM, Bouzeghrane Y, et al. Transformation and articulation of clinical data to understand students' clinical reasoning: a scoping review. *BMC Med Educ*. 2025 Jan 12;25(1):52. doi:10.1186/s12909-025-06644-7.
12. E B, S R, M A, Mn K, L S, W S, et al. Large language models improve clinical decision making of medical students through patient simulation and structured feedback: a randomized controlled trial. *BMC Med Educ*. 2024 Nov 28;24(1). doi:10.1186/s12909-024-06399-7.
13. Kononowicz AA, Woodham LA, Edelbring S, Stathakarou N, Davies D, Saxena N, et al. Virtual patient simulations in health professions education: systematic review and meta-analysis by the digital health education collaboration. *J Med Internet Res*. 2019 Jul 2;21(7):e14676. doi:10.2196/14676.
14. Çiçek FE, Ülker M, Özer M, Kiyak YS. ChatGPT versus expert feedback on clinical reasoning questions and their effect on learning: a randomized controlled trial. *Postgrad Med J*. 2025 Apr 22;101(1195):458–63. doi:10.1093/postmj/qgae170.

15. Rothlind E, Fors U, Salminen H, Wändell P, Ekblad S. Virtual patients reflecting the clinical reality of primary care – a useful tool to improve cultural competence. *BMC Med Educ*. 2021 May 11;21(1):270. doi:10.1186/s12909-021-02701-z
16. Chan CKY, Hu W. Students' voices on generative AI: perceptions, benefits, and challenges in higher education. *Int J Educ Technol High Educ*. 2023 Jul 17;20(1):43. doi:10.1186/s41239-023-00411-8
17. Tu T, Schaekermann M, Palepu A, Saab K, Freyberg J, Tanno R, et al. Towards conversational diagnostic artificial intelligence. *Nature*. 2025 Jun;642(8067):442–50. doi:10.1038/s41586-025-08866-7.
18. Alqaissi N, Qtait M. Knowledge, attitudes, practices, and barriers regarding the integration of artificial intelligence in nursing and health sciences education: a systematic review. *SAGE OPEN Nurs*. 2025;11:23779608251374185. doi:10.1177/23779608251374185.
19. Zhang B, Liu X, Wang Y, Zhou L, Xie Q, Wang B. Human or LLM as standardized patients? A comparative study for medical education. *arXiv [Preprint]*. 2025 Nov 12 [revised 2026 Mar 23; cited 2026 Jun 3]. Available from: <https://arxiv.org/abs/2511.14783v3>. arXiv:2511.14783v3.